

Turki Alzaid

Computer Science Student | Software Engineering

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Computer Science student at Loyola Marymount University building mobile, desktop, and game-systems software with AI coding agents. Defines requirements and architecture, directs and reviews implementation, debugs complex behavior, and verifies work with automated tests.

EDUCATION

Loyola Marymount University - Frank R. Seaver College of Science and Engineering
Bachelor of Science in Computer Science | Started Fall 2026

Expected 2030

Relevant study: Software engineering, programming, calculus, web fundamentals, SDLC, and requirements engineering

EXPERIENCE

Aida Designs - Designer's Assistant Intern

Los Angeles, CA | June 2024

- Supported architectural designers through blueprint reading, AutoCAD, Excel, site visits, client communication, team meetings, and supplier coordination.
- Worked with pricing forms, project constraints, client expectations, budgeting, billing, and custom-manufacturing coordination.

SELECTED PROJECTS

MacroMaxx - Native iPhone Nutrition Tracker

Product and Technical Lead | AI-assisted development

- Defined and directed an offline-first SwiftUI app with the full USDA Branded Foods database on device (SQLite), barcode lookup through Open Food Facts, HealthKit weigh-ins, a WidgetKit widget, and 1,054 restaurant items parsed from chains' published nutrition PDFs.
- Specified a Nutri-Score 2023 port and validated it against Open Food Facts' published grades (184 of 185 matched); 370 core tests passing. Shipped 17 builds to a personal iPhone with automated re-signing and hash-verified data backups before each install.

E.V.I.E. - Voice-First AI Assistant for Windows and macOS

Project Lead | AI-assisted development

- Defined architecture for a Windows desktop assistant using Tauri 2, React, TypeScript, Python, Rust, Whisper, Piper TTS, Claude, Ollama, Graphiti, and Neo4j.
- Directed and reviewed a typed local protocol, event-driven overlay, process lifecycle controls, system status/shutdown, and orphan-process prevention; validated 136/136 backend tests and four Rust tests.
- Rebuilt it as a dependency-free native macOS app in Swift (about 5,400 lines) with a global shortcut, on-demand dictation, approval gates for file changes, and editable reusable skills.

ClassPath - Campus Scheduling and Navigation App

Product and Technical Lead | AI-assisted development

- Defined requirements and directed implementation of graph-based campus routing, schedule timing, Today/Schedule screens, class editing, and local SQLite persistence.
- Reviewed routing validation, hash-approved campus data packs, schedule storage, navigation guards, accessibility requirements, and browser-preview behavior; verified 375 passing automated tests.

NebulaOpen - Minecraft Navigation and Pathfinding

Technical Lead | AI-assisted development

- Directed debugging and review of Java pathfinding systems involving route anchors, ledges, corridors, terrain classification, collision-aware movement, and camera arbitration.
- Specified and validated camera snap-back fixes and regression cases for walled passages, ledges, and bridges; verified with 886 passing tests.

Zero - Minecraft Client Prediction Systems

Systems Project | AI-assisted development

- Defined and reviewed authoritative-server versus predicted-client state handling, optimistic replay, state reconciliation, timeout behavior, and lifecycle queue clearing across Java/Kotlin modules.

SpotiHUD - Spotify Overlay for Minecraft and macOS

AI-assisted development

- Fabric mod with Spotify OAuth PKCE sign-in, a Tauri/Rust always-on-top desktop overlay, and a SwiftUI notch display whose equalizer uses a Core Audio tap and FFT.

TECHNICAL SKILLS

Languages: Swift, Java, TypeScript, JavaScript, Python, Rust

Frameworks and tools: SwiftUI, React, React Native, Expo, Tauri, Vite, Fabric, Gradle, Xcode, Git, PowerShell

Systems: Automated and regression testing, debugging, pathfinding, state reconciliation, process lifecycle management, local protocols, offline data, AI and voice integration

Other: SQLite, HealthKit, Neo4j, Graphiti, Whisper, Piper TTS, Ollama, AutoCAD, Microsoft Excel